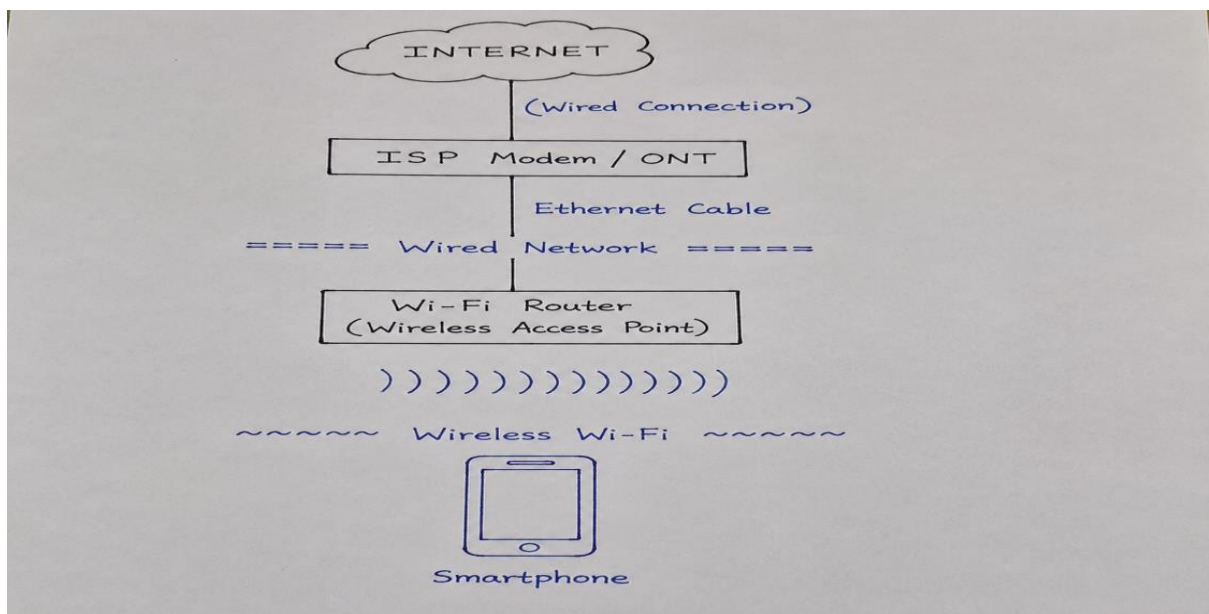


1. List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Wi-Fi Standard	Common Name	Key Difference
802.11n	Wi-Fi 4	Operates on 2.4 GHz bands with speed of up to 600 Mbps.
802.11ac	Wi-Fi 5	Operates only on the 5 GHz band and supports speed of up to 6.9 Gbps, making it faster than 802.11n.
802.11ax	Wi-Fi 6	Operates on 2.4 GHz and 5 GHz and supports speed of up to 9.6 Gbps, while also improving performance in crowded networks.

2. Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labelling the wireless and wired parts of the network.



3. Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.

SSID (Wi-Fi Name)	Signal Strength	Security Type
Airtel_Tops2	Excellent (86%)	WPA-2 Personal
ROHAN's Galaxy S22+	Excellent (90%)	WPA-2 Personal
MANPRISH	Excellent (85%)	WPA-2 Personal

- I would choose ROHAN'S Galaxy S22+ because it has the strongest strength (90%) which provides faster and more stable internet and a strong signal combined with better security offers overall best Wi-Fi experience.

4. Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

- SSID hiding is Wi-Fi security feature where a wireless router does not broadcast its network name (SSID). This means the Wi-Fi network does not appear in the list of available networks on nearby devices. To connect, users must enter the SSID and password.
- **Advantage**
- Reduces casual visibility which means network is not visible to people scanning for nearby Wi-Fi, which may discourage casual users from trying to connect.
- **Disadvantage**
- It does not have a strong security measure which means hidden SSIDs can still be detected using Wi-Fi scanning tools, so they do not provide real protection against attackers. They can also make it less convenient for legitimate users, who must manually enter the network name to connect.